

Street violence against women in urban spaces: A comparative analysis based on citizen perceptions and official data

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ABSTRACT

Violence in public spaces continues to be a critical challenge for citizen safety and well-being, with special focus on gender-based violence. This study compares the relationship between perceptions of insecurity and recorded incidents of violence across three cities with different socio-cultural and legislative contexts: Valencia (Spain), Toluca (Mexico), and Dublin (Ireland). For this purpose, data from social networks (X, formerly Twitter), participatory mapping results (mapathon), and police records were used, covering four different time scenarios ("working day, working night, holiday day, and holiday night").

The statistical analysis included Pearson correlation coefficients and linear regression models to assess the explanatory power of variables, such as harassment and negative mentions on social networks, for citizens' perceptions of insecurity. The findings show that harassment reported during mapathons is strongly and consistently correlated with perceptions of insecurity across scenarios.

On the other hand, the statistical association between social networks and other indicators is context-dependent and generally limited, varying according to temporal conditions and data availability. There is also a significant gap between official records and public perception, underscoring the need for more comprehensive approaches to data collection and analysis to strengthen public policymaking.

The findings highlight the usefulness of combining traditional methodologies with collaborative and digital sources, despite the limitations arising from the small data sample and the heterogeneity of urban contexts. This study underlines the importance of strengthening data collection and standardisation at a multi-city scale to advance the understanding and prevention of urban violence against women.

1. Introduction

Street violence (SV) refers to direct and visible forms of violence that negatively affect people's perception of safety (Saborío, 2019). This type of violence includes acts such as robbery, physical assault, and harassment in public spaces (Robles, 2014), and its impact on perceptions of insecurity is significant, as it generates fear, anxiety, and other emotional responses that affect the quality of life of urban dwellers (Robles, 2014). Perceptions of insecurity are not only based on direct experiences, but also on the information people receive from their environment, including the media and social networks (Robles, 2014).

SV persists as a serious social problem that, in the digital age, is also manifested and analysed through platforms such as X (formerly

Twitter), where victims and the community express and respond to incidents of SV. Recent studies highlight that social networks make these problems visible, leading to greater social awareness and collective support (ElSherief, Belding, & Nguyen, 2017). While previous studies, such as Battisti, Kauppinen, and Rude (2024), have analysed the impact of digital campaigns on Twitter to raise public awareness and influence policy-making related to gender-based violence, there remain few studies on how data from these social media platforms can be integrated and correlated with other sources of reports of SV to build accurate analytical models.

In parallel, studies such as that of Gargiulo (Gargiulo et al., 2020) have combined qualitative and quantitative data to assess women's perception of safety in urban spaces. Their approach has been mainly

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based on interviews and traditional spatial data, without including information derived from social networks. Such studies have shown that multiple factors, including urban context and previous exposure to violent conditions, affect perceptions of insecurity. However, how spontaneously generated data from digital platforms can complement and enrich these traditional approaches is yet to be explored in depth.

This research addresses this gap by integrating information from social networks, collaborative perception data (mapathon), and police records of assaults, intending to build a more robust and contextualised model of SV. Through the analysis of geolocated tweets and a participatory *mapathon*, this research enables us to assess the relationships among social media mentions, citizens' perceptions of insecurity, and police records in public space across different temporal scenarios. This multidimensional approach allows not only to compare perceptions of insecurity across cities and time slots, but also to assess the extent to which social networks can serve as a complementary indicator of urban insecurity. Based on these aspects, this article addresses the following research questions: (1) To what extent do participatory maps of harassment (H) and perceived insecurity (IP) converge or diverge from official police reports (PS) in different time scenarios in Valencia and Toluca? (2) To what extent can geolocated negative tweets related to street violence and harassment (X) complement these sources as indicators of urban insecurity in Valencia, Toluca, and Dublin, despite data and API limitations? (3) How do differences in the urban and institutional context, and in the availability of gender-disaggregated data, influence the usefulness and biases of each data source as an indicator of street violence against women?

This analysis forms part of a broader research framework that integrates quantitative and qualitative data. The article focuses on the perception-related components of the project dataset. These components are integrated with official records through quantitative geospatial and statistical analyses, and the datasets are available in Zenodo. (Anquela et al., 2025), while the detailed description of the dataset is published in Data in Brief (Hernández, Anquela, Martín, Martínez, & Fernández, 2025). These processes are carried out in accordance with the FAIR (Findable, Accessible, Interoperable, Reusable) principles (Wilkinson, Dumontier, Aalbersberg, et al., 2016), promoting transparency and reusability of the information to facilitate its understanding and applicability in future studies.

The main contribution of this study lies in combining geospatial analysis methodologies with social network data and collaborative mapping, thereby providing a more dynamic and contextualised view of perceptions of insecurity. Moreover, its findings have direct implications for the design of public policies, as they can guide urban violence prevention strategies by identifying spatio-temporal patterns of risk and optimising the allocation of security resources in public spaces.

2. Methodology

This study aims to analyse the influence of different data sources, including social networks, collaborative mapathons, and police records, on street violence against women in three cities with different urban characteristics and data contexts: Valencia (Spain), Toluca (Mexico), and Dublin (Ireland).

These three cities were selected to compare perceptions of insecurity across contexts with different legislative frameworks and levels of technological development, while also addressing data availability. European open data regulations govern both Ireland and Spain. Still, there are differences in the practical implementation of these policies across countries, which provide an opportunity to analyse a diversity of approaches within the European Union. For its part, Toluca brings the perspective of a country with high rates of gender-based violence and has significant limitations regarding the availability of official open data. While in European contexts an institutional infrastructure has established solid regulations that actively promote the publication, updating, and free access to government datasets, in Toluca, public

policies still do not guarantee openness, interoperability, or data reuse. This provides a broader and more robust comparison by examining the influence of sociocultural and accessibility factors on citizens' perceptions of insecurity.

2.1. Urban context and data sources

Valencia, located on the Mediterranean coast, is the third-largest city in Spain, with a population of approximately 830,000 and a surface area of 135 km². Its urban structure, marked by wide open spaces and a Mediterranean climate, provides an ideal context for analyzing the dynamics of insecurity perception in urban areas. The data from Valencia primarily comes from a collaborative mapathon and a set of negative tweets obtained via web scraping. Data from calls to the local police in Valencia (092) were used.

Toluca, capital of the Central State of Mexico, has a population of approximately 910,000 inhabitants and is an important economic and industrial center. In this case, the data includes a broader collaborative mapathon, negative tweet data, and emergency reports derived from 911 calls, providing a more diverse and quantitatively robust picture than in Valencia.

Dublin, Ireland's capital city, has a population of approximately 1,284,000. In this case, the harassment data was obtained from police reports that include a combination of urban safety-related events and geolocated negative tweets.

The variables used in this study were structured into three main categories, obtained through different methodologies in each of the cities analysed: Valencia, Toluca, and Dublin.

A. Social media data (X):

To analyse perceptions of street violence in Valencia, Toluca, and Dublin, posts were collected via scraping on the X platform. Although the restrictions of social network APIs make it challenging to obtain data, it is possible to overcome these restrictions by applying robust web scraping techniques, provided ethical, legal, and technical principles are respected (Harrell, Cruickshank, & Master, 2024). This approach allows us to access valuable information for analysis. All code was written in Python. The pandas, numpy, matplotlib, and seaborn libraries were used for initial data manipulation and visualization. Geolocated posts containing explicit references to perceptions of insecurity or street harassment events were selected based on a sentiment analysis process applied to all tweets collected within the defined spatiotemporal parameters, using a pre-trained model that automatically identifies negative emotional content. Although the specific algorithm is embedded in the Python environment, it relies on lexical polarity and contextual embeddings to classify posts.

For the city of Valencia, the dataset contains 18,472 records and eight variables. The variables used are listed in Table 1.

Subsequently, geolocation data (latitude and longitude) were ensured to be in the correct format and converted into usable geospatial points for analysis. The geolocation data were checked, cleaned, and converted to a geospatial format using GeoPandas.

The vectorisation of the texts was carried out using the TF-IDF technique (Das, K, & Alphonse, 2023; Gaydhani, Doma, Kendre, & Bhagwat, 2018) with the scikit-learn library, so that each word is represented numerically by its frequency and overall relevance. Following

Table 1
Variables retrieved from X (Twitter) scraping.

User	Name of the user who posted the tweet.
TimeStamps	Timestamp of the tweet.
Tweets	Text of the tweet.
Replies	Number of replies to the tweet.
reTweets	Number of retweets.
Likes	Number of "likes".
Latitude	Latitude coordinate of the tweet.
Longitude	Longitude coordinate of the tweet.

this, sentiment analysis was performed using the TextBlob library, which assigns a polarity (positive, negative, or neutral) to each post. This workflow combines the statistical weight of the terms with the emotional evaluation of the texts (Alemán Viteri, 2021).

In the next stage, advanced pre-processing, machine learning, and artificial intelligence techniques were applied to classify posts based on their content and sentiment, with a special focus on those that reflected harassment and insecurity. The classification was carried out using a CNN-LSTM model, trained on a manually labelled dataset to capture linguistic nuances related to risk perception and violence. To address class imbalance, oversampling techniques such as SMOTE (Synthetic Minority Over-sampling Technique) (Chawla, Bowyer, Hall, & Kegelmeyer, 2002) were applied during training. Additionally, TF-IDF vectorisation was retained as a textual representation strategy.

The CNN-LSTM model demonstrated strong performance in detecting negative posts, as evaluated by classification metrics such as precision, recall, and F1-score. These metrics provide a robust framework for assessing the model's ability to distinguish tweets correctly. Despite the limited volume of labelled tweets, temporal patterns and correlations were explored. Fig. 1 shows the training and validation loss and accuracy curves, further supporting the model's convergence and generalization over the training epochs. The same CNN-LSTM architecture and training procedure were applied to Spanish tweets from Valencia and Toluca, while a similar model was trained with English tweets from Dublin. Due to space limitations, we present detailed metrics for Valencia, although it should be noted that performance in Toluca and Dublin was of a similar magnitude (Table 2).

The sample of posts covers the dates 2022-10-07 to 2023-08-20, yielding 191 posts with negative sentiment toward Valencia, 118 toward Toluca, and 946 toward Dublin. This pattern is consistent with previous studies showing that posts with explicit geographic information about crimes are rare at the city level and that users who share their location are less likely to post content linked to criminal acts (Prieto Curiel, Cresci, Muntean, & Bishop, 2020).

To facilitate a meaningful comparison across the three study areas, Table 3 presents the volume of geolocated data relative to the urban population. The data collection process was restricted exclusively to geolocated posts at the source.

Given the varying densities of geolocated negative tweets across cities, different visualization strategies were employed to represent their spatial patterns best; in cities with higher tweet volumes, such as Valencia and Dublin, heatmaps or kernel density estimates were used to illustrate broader spatial concentrations. In contrast, for Toluca, with its extremely sparse yet highly concentrated negative tweets, a choropleth map showing raw counts and a normalized index per spatial unit was

Table 2

Classification report of the CNN-LSTM model in Valencia.

Class	Precision	Recall	F1-Score	Support
0	1.0	0.792	0.884	24
1	0.8	1.0	0.889	20
Accuracy	0.886			
Macro avg	0.9	0.896	0.896	44
Weighted avg	0.909	0.886	0.886	44

Table 3

Data volume and density by city.

City	Population (approx.)	Total Geolocated Tweets (X)	Negative Tweets Analyzed	Negative Tweets per 100 k inhabitants
Valencia	830,000	18,472	191	23.01
Toluca	910,000	307	118	12.96
Dublin	1,284,000	31,601	946	73.67

chosen to highlight specific areas of concern.

The DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm detected regions with many negative tweets. It first identifies "core points" with at least a minimum number of nearby neighbors (minPts) within a ϵ radius. This radius remained constant across cities for consistency. The algorithm expands clusters by identifying all tweets within the ϵ -radius of each core point. Points with enough neighbors join a cluster; others are labelled as noise. DBSCAN assumes no cluster shape, uses range queries, and is well-suited to geospatial data. It groups nearby tweets robustly and excludes outliers, mirroring real spatial distributions (Schubert, Sander, Ester, Kriegel, & Xu, 2017).

The process described above is shown in Fig. 2 for Dublin and in Fig. 3 for Valencia.

Fig. 4 shows a spatial clustering analysis of tweets with negative sentiment in the city of Valencia, carried out using the DBSCAN algorithm. The graph shows several points scattered on a two-dimensional plane, where the horizontal axis (x) corresponds to longitude and the vertical axis (y) to latitude.

The points are colored according to their membership in different clusters identified by the DBSCAN algorithm. In particular, the following colors and cluster labels are used:

- Cluster -1 (red dots): Represents the points considered as noise by the DBSCAN algorithm. These points do not belong to any significant cluster according to the defined parameters (ϵ and MinPts).

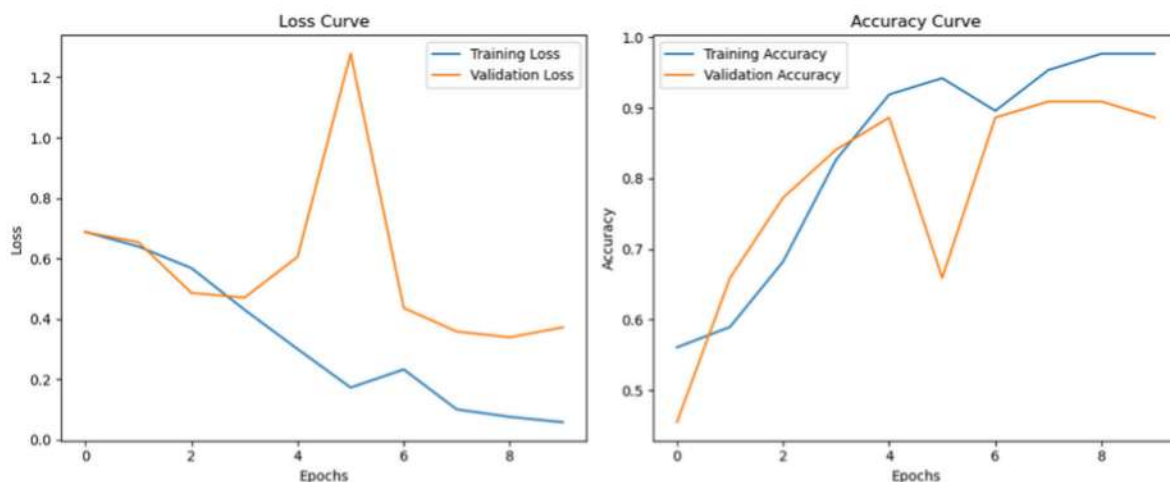


Fig. 1. Loss and accuracy curves of the CNN-LSTM Model.

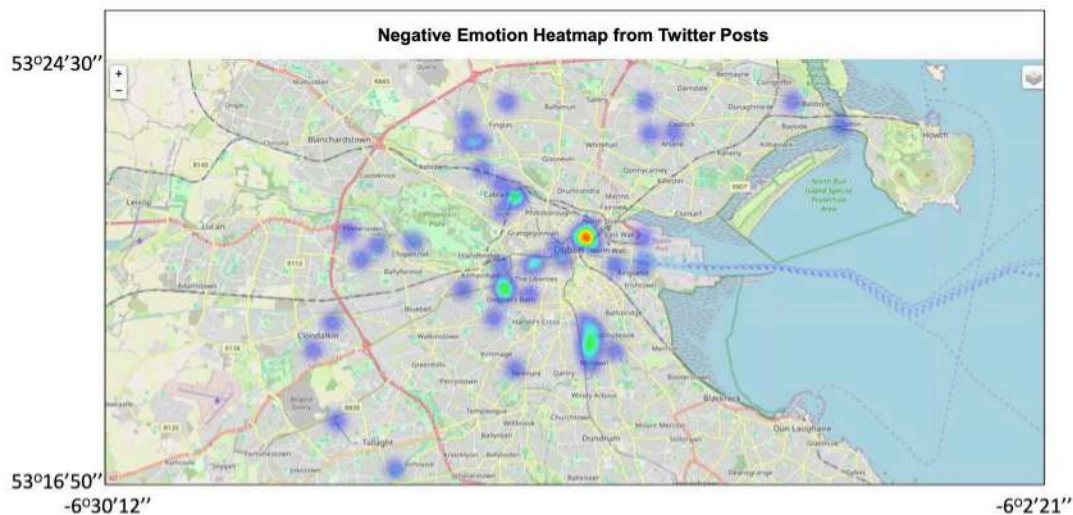


Fig. 2. Heat map of X's post with negative sentiments in Dublin. Source: authors' elaboration from X (Twitter) scraping.

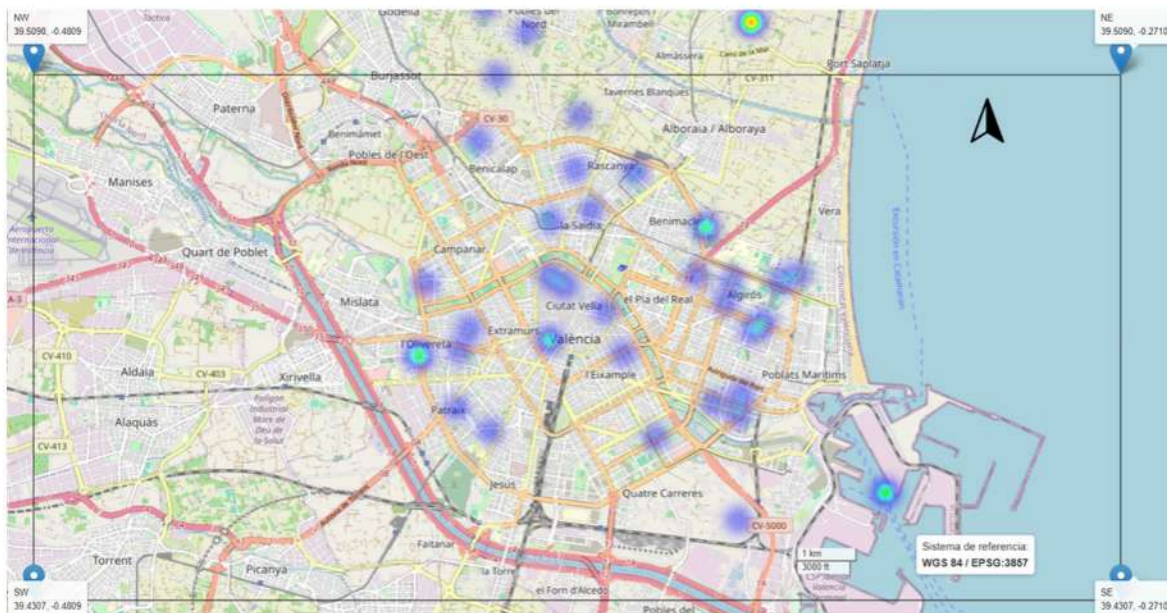


Fig. 3. Heat map of X's post with negative sentiments in Valencia. Source: authors' elaboration from X (Twitter) scraping.

- Clusters 0, 1, 3, 4, 6 (different colored dots): Each of these colors represents a distinct cluster, where the points within each cluster are geographically close to each other and meet the set density criteria.

In simpler terms, the graph shows how tweets expressing fear are clustered in some regions of Valencia. This suggests that, in these areas, people perceive greater insecurity, particularly related to gender-based violence.

Choropleth maps for Toluca (Fig. 5) and Valencia (Fig. 6) were generated to display the raw number of negative tweets per spatial unit. This approach was chosen to effectively highlight specific areas of concern within a dataset characterized by low overall volume but high local intensity.

B. Data on citizens' perception of insecurity on the streets and harassment (mapathon):

Citizen science techniques were used to obtain information on perceptions of insecurity in public spaces and situations of harassment. This qualitative data was obtained through face-to-face events organised as part of a collaborative mapathon. A mapathon allows citizens, regardless

of their professional background in cartography, to actively participate and has expanded the scope and accuracy of geospatial data, enabling applications across diverse areas such as environmental management and urban development (Claramunt & Lotfian, 2023).

This process was carried out in the cities of Toluca and Valencia. During these events, participants were instructed to map areas perceived as unsafe or associated with harassment using an application developed for this purpose. This web application combines digital tools, such as OpenStreetMap, with specific questionnaires designed to collect details on perceptions of public space. In Toluca, the activities took place at several venues at the Autonomous University of the State of Mexico, while in Valencia, they were held in person and online at the Polytechnic University of Valencia (UPV).

Participants classified the geographical locations into three categories:

- Harassment (H) (related to behavioral incidents, physical or verbal aggression).

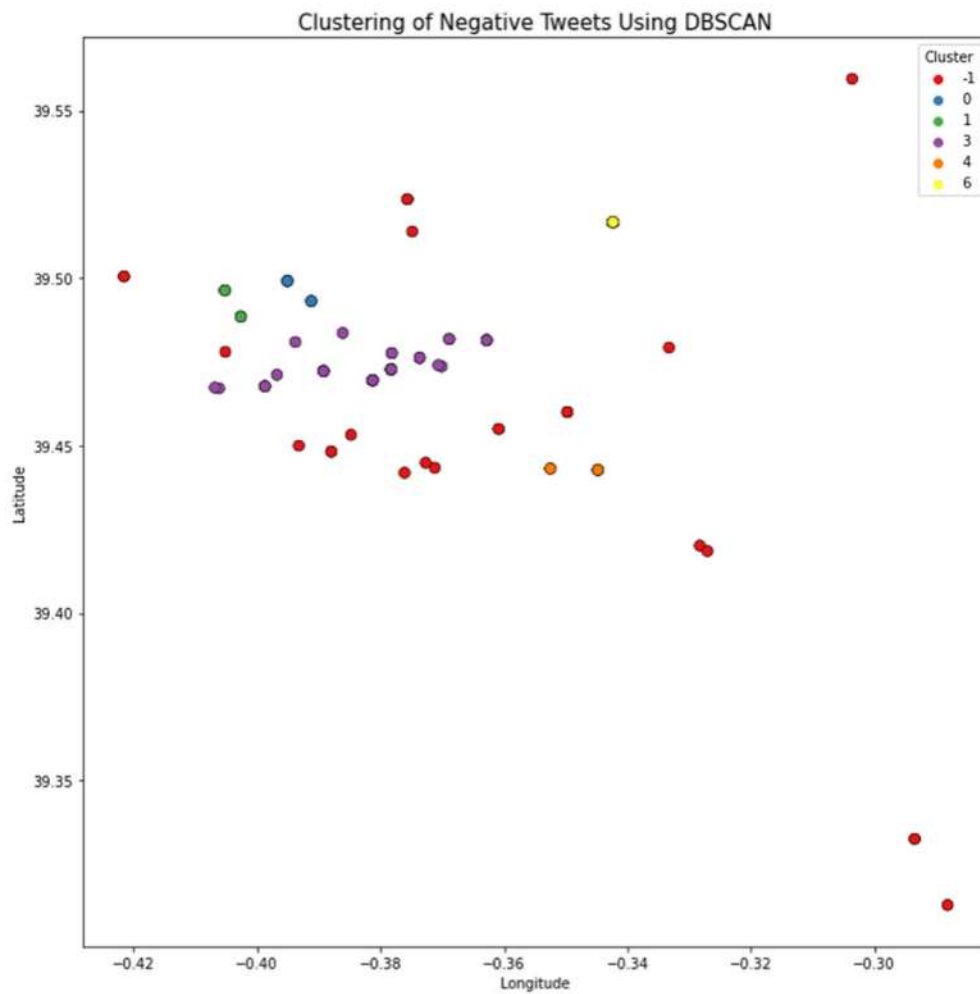


Fig. 4. Technical analysis of clustering of negative Tweets in Valencia using DBSCAN.

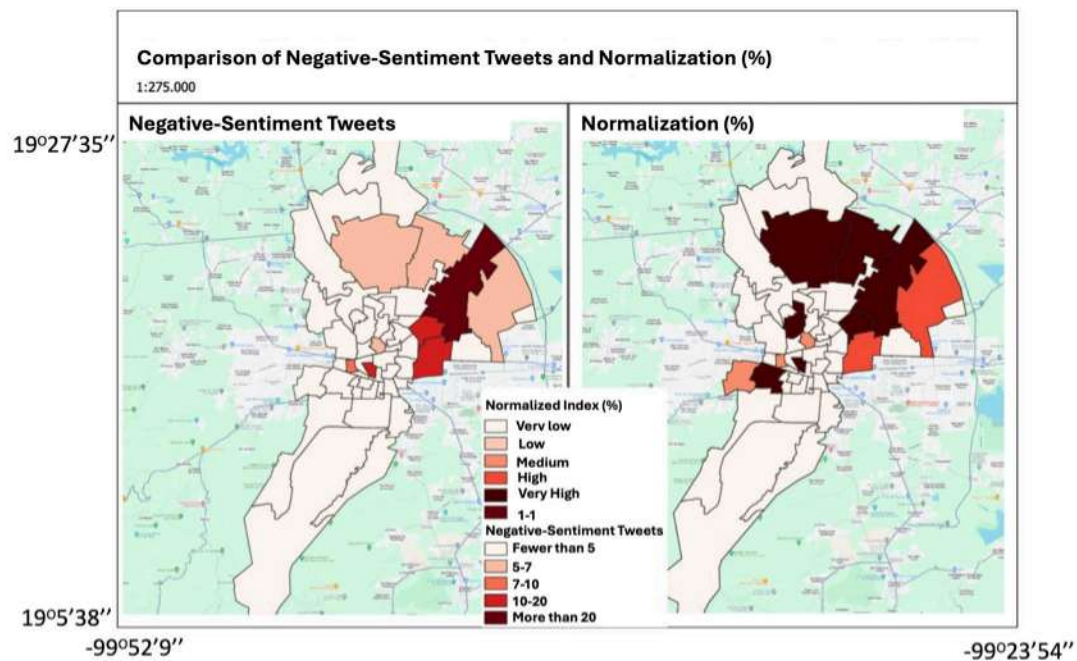


Fig. 5. Spatial distribution of negative-sentiment tweets in Toluca: raw counts and normalized index.

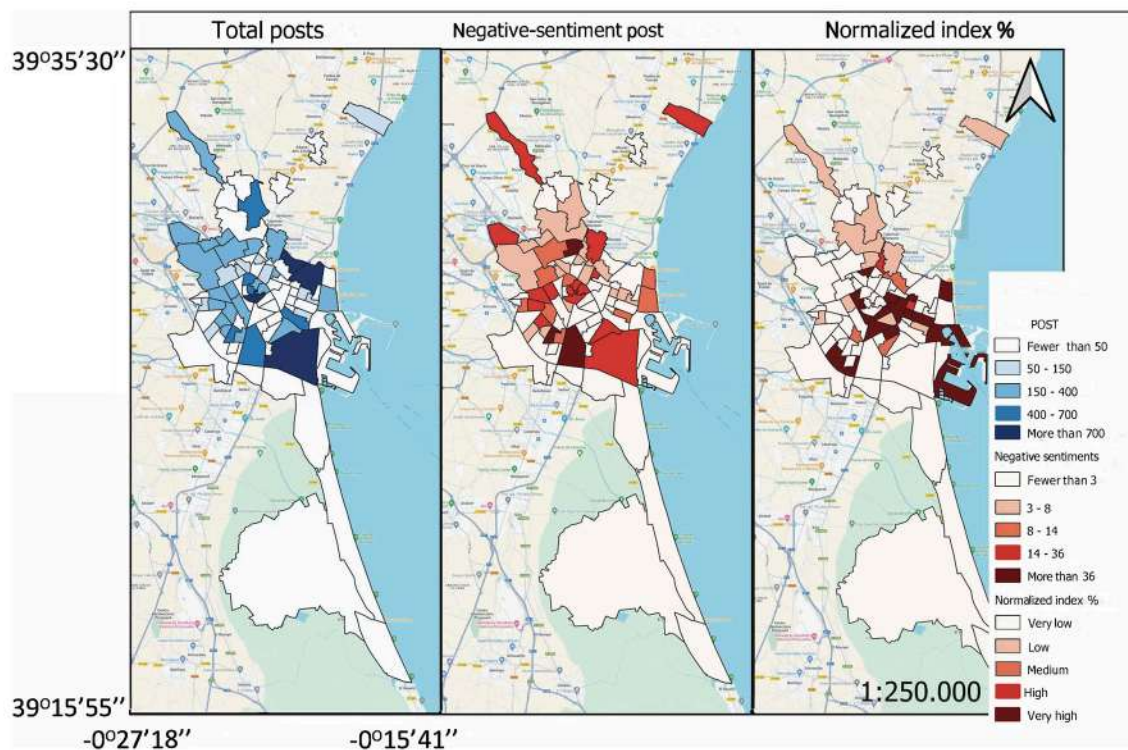


Fig. 6. Spatial distribution of negative-sentiment tweets in Valencia: total post, raw counts and normalized index.

- Citizens' perception of insecurity (IP) (areas perceived as risky due to factors such as lighting, pedestrian traffic density, etc.).
- Mixed areas (citizens simultaneously marked this geographic point to simultaneously report areas that combined perceptions of insecurity and harassment events).

The Mapathon events achieved significant engagement in both cities, with participant profiles heavily weighted toward the study's target demographic. In Valencia, 72 individuals participated, of whom 59 were women (81.9%), 12 were men, and 1 preferred not to state their gender. In Toluca, participation was even more extensive, reaching 668 individuals: 654 women (97.9%), 7 men, and 7 who preferred not to answer. This overwhelming majority of female participants, particularly in the Toluca case, ensures that the spatial data collected through the Mapathons predominantly reflects women's lived experiences and perceptions of safety, directly addressing the gender-specific focus of this research.

Data reported in mixed areas were counted in both harassment (H) and citizens' perception of insecurity (IP).

Although harassment (H) and perceived insecurity (IP) were collected in the same participatory mapping process and could be reported simultaneously, they are conceptually distinct constructs. Harassment (H) is a prevalent form of violence against women (VAW) that refers to experienced or witnessed events of physical, verbal, or behavioral aggression in public. It is described as a situation in which an unknown person makes comments, gestures, or actions (such as whistling, sexual gestures, touching, or stalking) in a public place, directed at the victim, and without their consent (Vázquez-González & Ferrer-Pérez, 2025). In contrast, perceived insecurity (IP) captures anticipatory fear or risk assessment in locations that may not be linked to a specific incident (e.g., poor lighting, low pedestrian traffic, or an isolated urban morphology). The IP is a broader concept that is understood as a network of representations, discourses, emotions, and actions related to current dangers in the social environment (Almanza Avendaño, Romero-Mendoza, & Gómez San Luis, 2022). Participants could therefore (i) map locations with harassment but without a strong, persistent perception of

insecurity, (ii) map places perceived as unsafe despite never having experienced harassment there, and (iii) mark mixed locations where both conditions co-occurred. Mixed locations were included in both variables, which introduces some statistical dependence. Still, the coding scheme preserves the analytical distinction between event-based violence (H) and broader perceptions of insecurity (IP).

Fig. 7 shows the reports obtained in the mapathons held in the city of Valencia.

In Toluca, participation was broader (1819 points), resulting in a more complete dataset than in Valencia (117 points), where the sample was more limited.

Unlike in Valencia and Toluca, no participatory mapping was held in Dublin within the timeframe and scope of the project funded by the Valencian Regional Government. Mapathons are a demanding methodology that requires strong pre-existing collaborations with local universities, public administrations, and community organisations to recruit and train participants, secure appropriate venues, and manage the ethical and logistical aspects of the activity. In Valencia and Toluca, the research team could build on long-standing institutional partnerships that facilitated the joint design and implementation of the events. In Dublin, by contrast, collaboration with Technological University Dublin was established after the original project proposal and budget had been defined, which limited the resources and time available to organise a mapathon of comparable depth and scale. For this reason, the joint work in Dublin focused on the collection and analysis of police records (PS) and geolocated tweets (X), providing a first quantitative approximation to patterns of urban insecurity. Extending the participatory component to Dublin through a future mapathon is envisaged as the next step in this line of research.

C. Police records (PS):

The data source used is detailed below.

- Valencia: The source of data used is the calls to the emergency number of the local police of Valencia, 092. We have worked with information on crimes against sexual freedom and crimes against persons in the case of assaults and gender violence. We have used

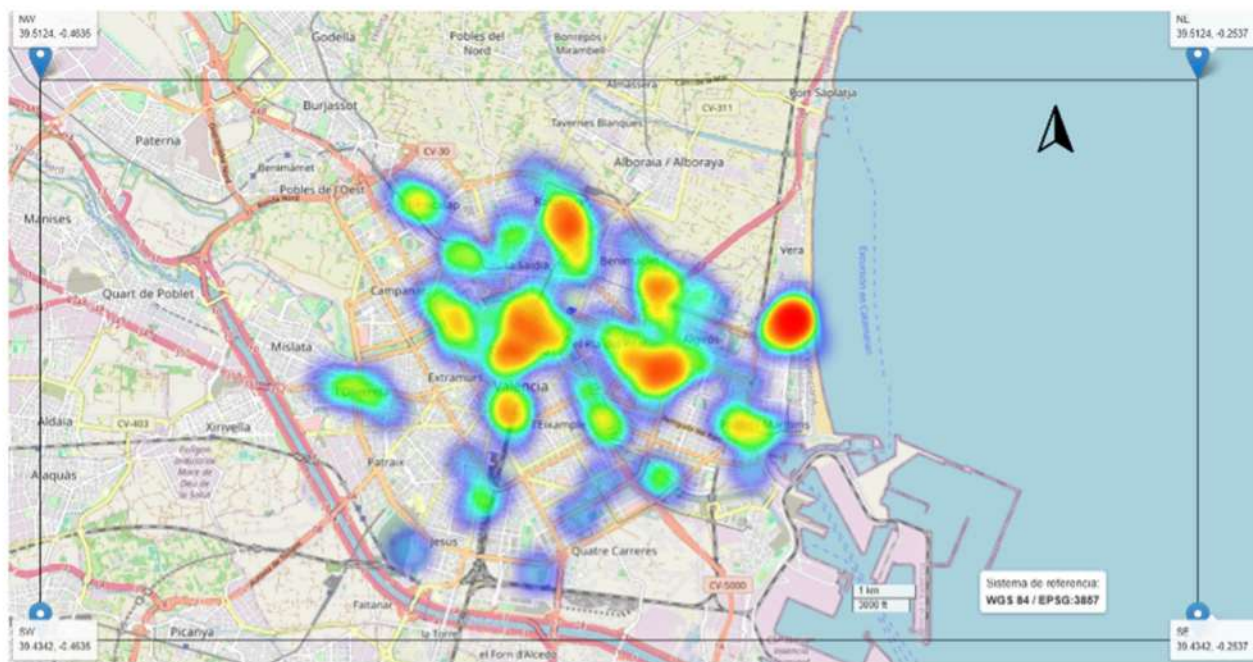


Fig. 7. Heat map with episodes of harassment and perceptions of insecurity. Source: authors' elaboration from mapathon data.

information from the period 7/6/18 to 30/05/24, resulting in 17,299 records.

- **Toluca:** The data source is the 911 emergency system: 25124 points corresponding to 911 calls administered by the Secretariat of Security of the State of Mexico in the period from 01/01/2020 to 31/05/2023 have been used. The total number of records in the dataset includes telephone calls classified according to the following categories: sexual harassment, homicide, rape, violence against women, intimate partner violence, and domestic violence.
- **Dublin:** The data source is assaults and police records aggregated by spatial unit and obtained from official open-data sources. The study includes a combination of events related to urban violence and risk behaviours in public space. The information analysed corresponds to the period 01/01/2018 to 01/01/2023, and 135,361 records were obtained.

Unlike the participatory data collected in Valencia and Toluca, the Dublin police dataset is not gender-disaggregated and includes a broad range of public-space incidents. Therefore, it cannot be interpreted as a direct measure of violence against women (VAW), but rather as a proxy for reported urban violence patterns occurring in public space. Consequently, the Dublin case is analysed as a contextual comparison within a different measurement regime rather than as validation of gender-specific street harassment dynamics.

2.2. Ethical considerations

All social media data used in this study were obtained from publicly accessible posts and analysed in aggregate, without attempting to identify individual users. For the mapathons, participants received detailed information about the study's objectives. They gave their informed consent for the anonymous use of their responses for research purposes, in accordance with the ethical guidelines applicable at the Polytechnic University of Valencia. Police records were used in an aggregated and anonymous manner.

2.3. Statistical analysis

The research on street violence against women was structured

through multivariate statistical analysis, using Pearson correlation and linear regression models to assess the statistical dependence and explanatory capacity of the variables, including the three input data. This analysis was applied in three cities (Valencia, Toluca, and Dublin) and in four different time scenarios: working day (s0), working night (s1), public holiday (s2), and public holiday night (s3). The statistical tests, detailed in the following section, allow us to infer the extent to which data from social networks (X) and reports of harassment (H) reflect or explain citizens' perceived insecurity (IP) in public space.

Calculation of correlations: Pearson's correlation coefficient r was used to assess the linear association between variables. It allows quantification of the relationship between two variables on a scale from -1 to 1 , where values close to 1 or -1 indicate a strong positive or negative linear relationship, respectively. For values close to 0 , the association is weak or non-existent (Cohen, West, & Aiken, 2014).

Regression models: Simple linear regression was applied to assess the explanatory association of independent variables (e.g., negative tweets (X) and harassment (H)) on the dependent variable, citizens' perceived insecurity (IP). The models were evaluated using the coefficient of determination (R^2), which measures the percentage of variance explained by the model (Cohen et al., 2014).

Data visualization: Scatter plots were used to analyse the relationship between the most relevant variables, selected based on the results of the correlation analysis ($r > |0.3|$). These visualisations were enriched with second-degree polynomial trend lines to capture non-linear patterns in the data and to avoid overfitting or underfitting (Cohen et al., 2014).

Differences in data quality and availability across the three cities influenced the study's methodological design. In Valencia, the limited number of tweets (X) and mapathon data (IP and H) reduced the representativeness of the sample and the robustness of the statistical analyses. In Toluca, incorporating police records (PS) enriched the study, though the volume of data from social networks (X) was also reduced. On the other hand, in Dublin, the absence of a mapathon (IP and H) and the exclusive reliance on police reports (PS) led to heterogeneous data sources, making direct comparisons between cities difficult.

These methodological differences were taken into consideration in the interpretation of the results and are considered a limitation of the study.

Pearson correlation coefficients and ordinary least squares regression models were estimated at the spatial-unit level (neighbourhood). We report two-tailed p -values for all correlation coefficients and regression parameters. Statistical significance is assessed at $p < 0.05$.

3. Results

In Dublin, the heatmap of negative tweets (Fig. 2) reveals a clearly non-uniform pattern, with higher densities of negative posts in and around the city centre and in some peripheral neighborhoods. In Toluca, the map of raw counts, together with the normalized index (Fig. 5), suggests a small set of spatial units with relatively more negative tweets. However, absolute numbers remain low due to the unique characteristics of its social media data. As detailed in the Methods section, Toluca exhibited a minimal total number of geolocated tweets (307), yet a disproportionately high percentage (38%) was classified as negative. This combination of extremely low overall volume and high concentration of negativity made a detailed, localized visualization particularly pertinent for Toluca.

1) Valencia

In Valencia, scenario s0 (working day) (Fig. 8) shows a strong positive correlation between harassment (H) and perceived insecurity (IP) ($r = 0.78, p < 0.001$). The linear regression model confirms this association ($\beta = 1.425, p < 0.001, R^2 = 0.60$). Approximately 60% of the variance in IP is explained by H in this scenario.

In the work night scenario (s1) (Fig. 9), a strong and statistically significant correlation is observed between negative tweets (X) and harassment (H) ($r = 0.66, p < 0.001, R^2 = 0.44$), suggesting that in this specific temporal context negative tweets partially capture the spatial distribution of reported harassment in the public nightlife environment. The corresponding linear regression model confirms this relationship, with the coefficient for negative tweets remaining highly significant ($p < 0.001$). However, the association between harassment (H) and citizens' perceived insecurity (IP), although still statistically significant, is more moderate than during the working day ($r = 0.5, p < 0.001, R^2 = 0.252$). This decrease in explanatory power compared with s0 indicates that additional contextual factors related to night-time urban dynamics or subjective risk perception may influence citizens' perceived insecurity during evening hours.

In the contexts of public holiday (s2) and festive night (s3) scenarios (Fig. 10) in Valencia, the association between harassment (H) and citizens' perceived insecurity (IP) remain strong and statistically significant. In s2 (holiday day), Pearson's correlation coefficients reach $r = 0.81$ ($p < 0.001$), with the linear regression model explaining approximately 66% of the variance ($R^2 = 0.664, \beta = 1.909, p < 0.001$). In s3 (holiday night), the relationship remains positive and significant, although

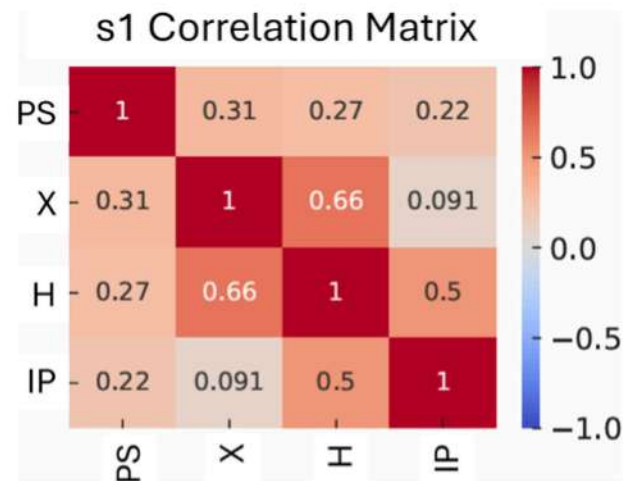


Fig. 9. Pearson's correlation in scenario s1.

somewhat weaker ($r = 0.59, p < 0.001; R^2 = 0.344, \beta = 1.575, p < 0.001$). These results reinforce the methodological interdependence between both variables, derived from the constructions process (IP and H). In contrast, the relationship between negative tweets (X) and perceived insecurity (IP) is weak and not statistically significant in either festive scenario (s2: $r = 0.1, p = 0.443; R^2 = 0.007$; s3: $r = 0.24, p = 0.379; R^2 = 0.009$) indicating that social network data (X) have very limited predictive capacity for perceived insecurity during holiday contexts.

Across the four temporal scenarios in Valencia, the correlation between harassment (H) and perceived insecurity (IP) ranges from $r = 0.50$ (s1) to $r = 0.81$ (s2), with R^2 values between 0.25 and 0.66. This consistently strong association suggests a substantial alignment between reported harassment and perceived insecurity within the participatory dataset. However, as discussed above, part of this explanatory power may be influenced by the inclusion of overlapping ("mixed") observations in both variables.

2) Toluca

In Toluca, data from tweets (X), mapathon (IP and H), and police records (PS) were analysed. In the workday scenario (s0), the Pearson correlation between police records (PS) and negative tweets (X) was moderate and statistically significant ($r = 0.28, p < 0.001; R^2 = 0.081$), indicating that tweets capture a limited but measurable share of reported incidents. At the same time, the relationship between harassment (H) and citizens' perceived insecurity (IP) was very strong $r = 0.90, p < 0.001; R^2 = 0.814$, indicating that harassment (H) explains a large proportion of variance in perceived insecurity (IP) in this context. The corresponding linear regression model confirms this finding ($\beta = 1.257, p < 0.001$). This result is consistent with the methodological structure of the citizens' perceived insecurity (IP) variable, which partially incorporates harassment (H) factors (Figs. 11 and 12).

In the working night (s1) scenario, negative tweets (X) showed a moderate to strong and statically significant correlation with police records (PS) ($r = 0.61, p < 0.001; R^2 = 0.375$), suggesting that, in this context, social network data (X) partially captures real-time reported violence (PS). However, the relationship between harassment (H) and citizens' perceived insecurity (IP), although still high ($r = 0.85, p < 0.001; R^2 = 0.753$), shows a slight reduction in explanatory power compared to the workday. The regression model confirms this association ($\beta = 1.302, p < 0.001$), which might indicate a marginal influence of other contextual factors on citizens' perceived insecurity (IP) during night-time hours (Fig. 13).

In the festive contexts of Toluca (day and night), the correlation between harassment (H) and citizens' perceived insecurity (IP) remains high and statistically significant (s2: $r = 0.92, p < 0.001; R^2 = 0.848$; s3:

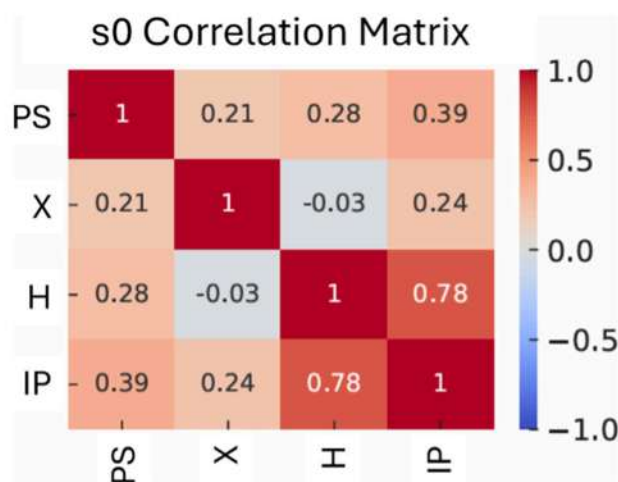


Fig. 8. Pearson's correlation in scenario s0.

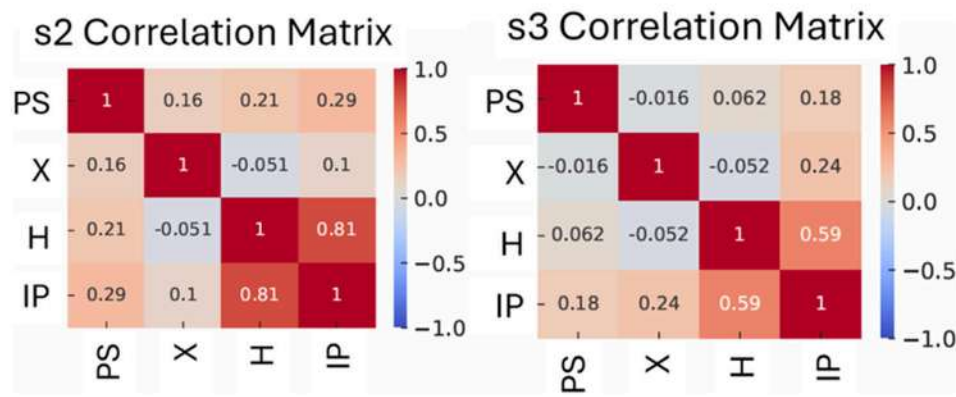


Fig. 10. Left: Pearson correlation in s2. Right: Pearson correlation in s3.

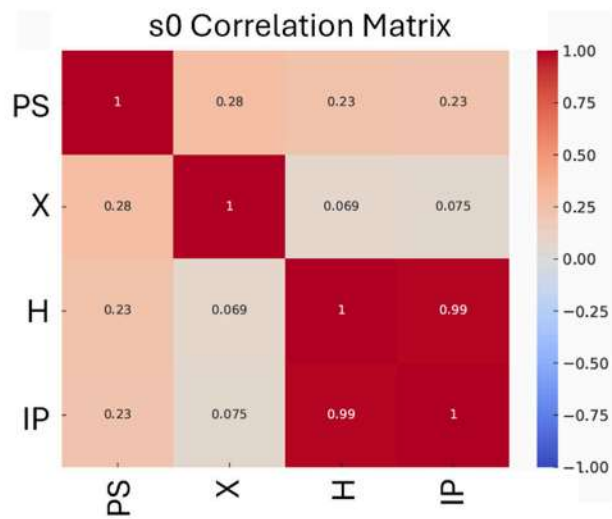


Fig. 11. Pearson's correlation in scenario s0.

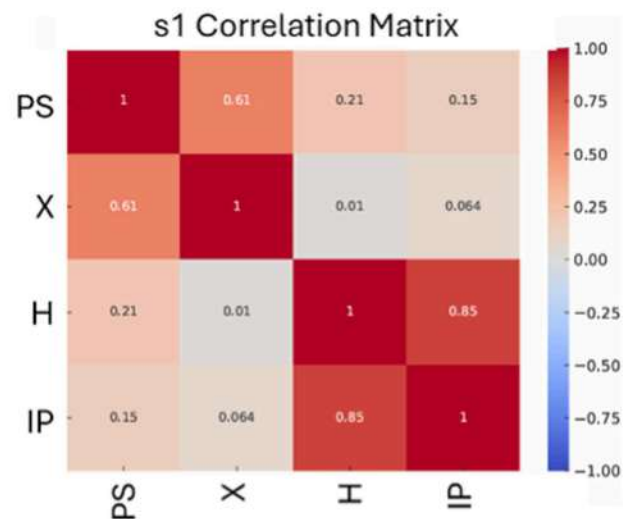
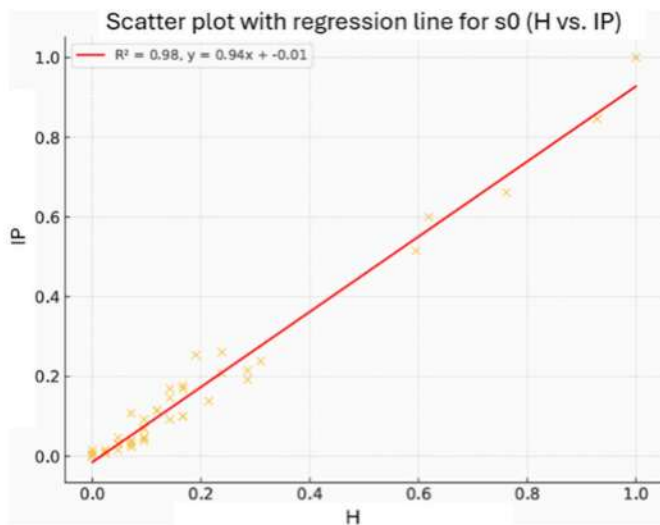


Fig. 13. Pearson correlation in s1.

Fig. 12. R^2 between harassment and insecurity in scenario s0.

$r = 0.918$, $p < 0.001$; $R^2 = 0.843$) reinforcing the methodological interdependence between both variables. In contrast, the correlations between police records (PS) and negative tweets (X) are weak and not statistically significant in these contexts (s2: $r = 0.04$, $p = 0.514$; s3: $r =$

0.07 , $p = 0.257$) suggesting that, in festive contexts, these external indicators have a limited capacity to reflect police-recorded incidents (Fig. 14).

3) Dublin

For the analysis of Dublin, data from negative tweets (X) and police reports (PS) were used. Fig. 15 presents the correlation for all scenarios in a single graph. Police records (PS) encompass a wide range of urban incidents, introducing additional heterogeneity into this variable. In the working day scenario (s0), a weak but statistically significant association is observed between negative tweets and police records ($r = 0.153$, $p = 0.008$; $R^2 = 0.024$) although the explained variance remains minimal. In the working night scenario (s1), the relationship is weak and not statistically significant ($r = 0.077$, $p = 0.184$; $R^2 = 0.006$). During the holiday day scenario (s2), a small positive association is observed but does not reach conventional levels of statistical significance ($r = 0.11$, $p = 0.057$; $R^2 = 0.012$). In the holiday night scenario (s3) the correlation becomes statistically significant ($r = 0.206$, $p < 0.001$; $R^2 = 0.042$), suggesting that negative tweets partially reflect police-recorded incidents in this context. However, even in this scenario, the explanatory power remains low, indicating that social network data provide only a limited proxy for officially recorded violence.

4) Robustness check: H-IP relationship excluding overlapping observations

To assess the extent to which the high correlations between harassment (H) and perceived insecurity (IP) are mechanically induced by overlapping ("mixed") classifications, we reconstructed both variables excluding all mixed observations. Because excluding overlapping points

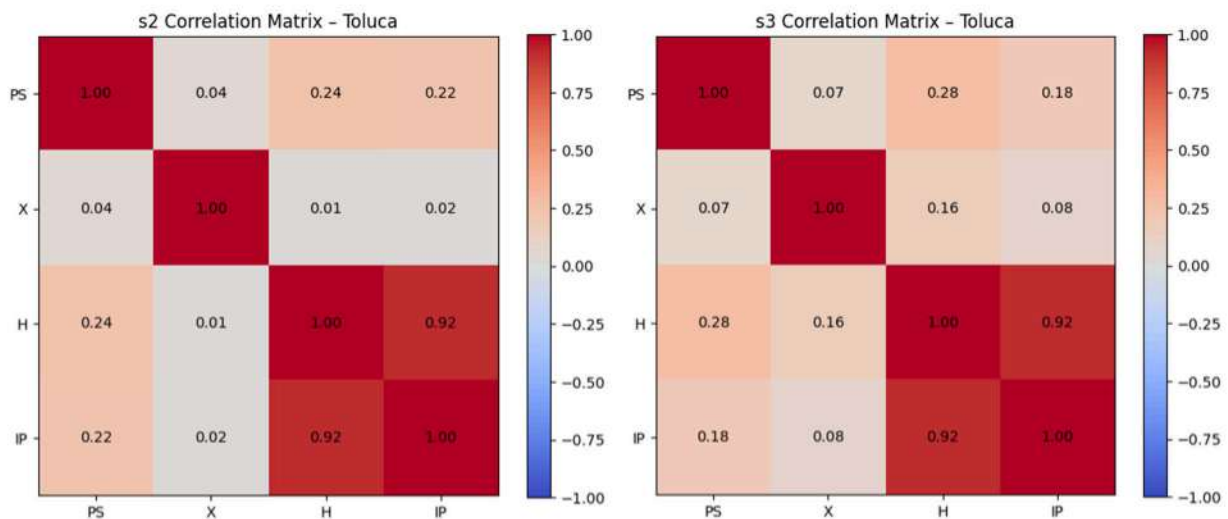


Fig. 14. Left: Pearson correlation in s2. Right: Pearson correlation in s3.



Fig. 15. Pearson's correlation in all scenarios.

at the scenario level substantially reduces the number of observations, this robustness analysis was conducted using aggregated totals across the four temporal contexts.

In Valencia ($n = 88$ neighborhoods), the original scenario-specific correlations between H and IP ranged from $r = 0.50$ to $r = 0.81$. After excluding overlapping observations and aggregating scenarios, the correlation decreased to $r = 0.41$ ($R^2 = 0.17$, $p < 0.001$). This substantial reduction in explanatory power indicates that a considerable portion of the original variance explained was structurally induced by the construction of the variables. Nevertheless, the persistence of a moderate and statistically significant association suggests that the spatial patterns of harassment and perceived insecurity are not purely mechanical.

In Toluca ($n = 48$ urban districts), scenario-specific correlations originally ranged between $r = 0.81$ and $r = 0.91$. After excluding overlapping observations and aggregating scenarios, the association decreases to $r = 0.52$ ($R^2 = 0.27$, $p < 0.001$). As in Valencia, the attenuation confirms that part of the initial explanatory strength was driven by construction effects. However, the remaining moderate and statistically significant relationship indicates substantive spatial

alignment beyond the mechanical overlap.

4) Comparative analysis between cities

Taking police reports (PS) as the primary reference, there are notable differences in the way mapathon and social media complement or diverge from these official records. In Dublin, where the mapathon was not conducted, police information is almost the only basis for comparison and shows limited correlations with available social media data. In Valencia, local police records are partially correlated with mapathon data and negative social media mentions, suggesting that although there is some correspondence, citizen perceptions do not always fully align with the events reported in emergency calls. Finally, in Toluca, the combination of 911, mapathon, and social media records allows observation of stronger internal associations between official information and citizen participation, allowing observation of stronger internal statistical associations within that data regime.

The results of this study provide a comprehensive interpretation of the relationships among social network data (X), citizens' perceived insecurity (IP), reports of harassment (H) from the mapathon, and police records (PS). This analysis confirms that harassment (H) is a key factor in perceptions of insecurity, whereas negative tweets (X) have limited, highly context-dependent explanatory power. However, limitations in data collection, particularly the availability of samples, pose a significant challenge to the generalisability of the results.

4. Discussion

The findings of this project suggest that while social networks are potentially useful as complementary sources, they are neither sufficiently representative nor reliable to serve as a primary basis for insecurity analysis. Furthermore, the so-called APICalypse – a term referring to the widespread restrictions on access to social media data via public APIs implemented after the Cambridge Analytica scandal in 2018 – has significantly limited the availability of structured digital data, making it difficult to retrieve information systematically (Bruns, 2019).

Since Twitter's switch from Twitter to X, the platform has undergone continuous changes to its web architecture, complicating web scraping. The site structure has become more dynamic and heavily reliant on JavaScript, which means that much of the content is loaded asynchronously or even hidden behind client-side rendering mechanisms, making it challenging to identify and extract data. In addition, HTML masking has increased, and more aggressive anti-bot measures are implemented, such as the insertion of session tokens, forcing continuous adaptation to a changing structure. Furthermore, many tweets lack geolocation, which makes working with this data more difficult. Moreover, social media

posts may refer to past events or incidents occurring at different times and locations than the moment of publication. This potential temporal and spatial mismatch between events and posts may attenuate observed correlations between official records and digital expressions of insecurity, particularly when geolocation metadata are sparse. As a result, even statistically significant associations should be interpreted cautiously, as temporal misalignment may reduce the observable strength of relationships between datasets.

Additionally, although some statistically significant correlations are observed between X and official or participatory indicators in specific scenarios (particularly in Toluca s1 and Dublin s3), their explanatory power remains low (R^2 generally below 0.05 in Dublin and moderate in Toluca s1). This indicates that social media data capture only a limited and context-dependent portion of formally recorded incidents and citizens' perceived insecurity.

The greatest challenge of this study lies in the small sample sizes, especially regarding negative tweets (X). In both Valencia and Toluca, the number of tweets (X) resulting from classifying posts by content and sentiment was very low, limiting the robustness of the conclusions that can be drawn from these data. This limitation is partly related to our methodological choices (restriction to geolocated tweets and focus on explicitly negative content). However, it also reflects structural properties of the platform: at the urban level, geolocated crime-related posts are intrinsically scarce, and users who share their location are less likely to post about crime than those who do not (Prieto Curiel et al., 2020). This sampling bias is particularly problematic in Valencia, where, in addition to the paucity of tweets (X), the mapathon data (IP and H) were also minimal, reducing the representativeness of the results. In Toluca, although the mapathon (IP and H) sample was larger than in Valencia, the social media data (X) remained too limited to support generalisable conclusions. Consequently, throughout the paper, we treat X as an exploratory and complementary indicator of online attention and concern rather than as a representative measure of women's experiences. A key challenge of this study stems from the heterogeneity of data sources across cities, driven by differences in legislative, technological, and information availability factors. While in Valencia (Spain) and Toluca (Mexico), data were available from the social network X, the mapathon, and emergency call records, in Dublin (Ireland), it was not possible to hold a participatory mapping event. This is a further limitation, as no participatory mapathon was conducted in Dublin due to the timing and resource constraints described in the Methods. As a result, the Dublin case relies only on PS and X, which restricts direct comparison with Valencia and Toluca for the participatory indicators (IP and H).

Additionally, the methodological linkage between the harassment and insecurity variables extracted from the mapathon introduces a bias in the correlation results, as expected. Since the insecurity variable partly captures the incidence of harassment (as both conditions can be reported at the same location), the high correlations between these variables reflect statistical overlap rather than an independent relationship.

The robustness analysis confirms this interpretation: although excluding overlapping observations substantially reduces the magnitude of the correlation, a statistically significant association remains in both cities. This suggests that the relationship between harassment and perceived insecurity is only partially driven by construction effects and retains substantive spatial coherence.

A notable aspect of the results is that comparatively stronger associations between social media data and official or participatory indicators tend to emerge during night scenarios. Although the magnitude of these effects varies across cities and remains moderate or weak in some cases, this pattern suggests that nocturnal contexts may reduce background social noise and produce comparatively clearer signals in social media data.

5. Conclusions

Overall, Pearson correlation analyses indicate that the association between harassment (H) and perceived insecurity (IP) in Valencia and Toluca is partially driven by the way both variables were constructed in the mapathon. The robustness analysis shows that although the explanatory power attenuates substantially when overlapping observations are excluded, a moderate and statistically significant association remains.

Negative tweet data (X), while showing moderate associations in specific contexts (e.g., working nights in Toluca and festive night in Dublin), do not exhibit consistent strong predictive ability across scenarios. In all cities, the explanatory power of X remains limited (R^2 generally below 0.05 in Dublin and moderate only in Toluca s1), indicating that social media data function better as contextual signals than as robust explanatory indicators. Consequently, it is concluded that while mapathon data (IP and H) and police reports (PS) provide statistically strong representations within their respective measurement frameworks, social networks should be used as complementary data rather than as a primary source for analyzing citizens' perceived insecurity (IP) or harassment (H).

Accordingly, the combination of a small, selective sample of geolocated tweets and a limited number of case cities means that our results should be interpreted with caution. They nevertheless highlight plausible mechanisms—such as under-reporting to the police, anticipatory fear shaped by urban form, and the amplification of specific incidents through news and social media—that structure the complex relationships between online discourse, citizen perceptions, and formally recorded violence.

Finally, we emphasize that the comparative component of this study is illustrative rather than conclusive: it highlights how different data ecosystems (open data regimes, citizen participation infrastructures, and social media usage) shape the observable relationships between violence against women and perceived insecurity, rather than providing definitive rankings of cities.

CRedit authorship contribution statement

Ana Belén Anquela: Writing – review & editing, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Ángel Martín:** Validation, Supervision, Methodology, Investigation. **Sandra Hernández:** Software, Methodology, Formal analysis, Data curation. **Eloina Coll:** Writing – review & editing. **Juan Vicente Chiva:** Writing – original draft, Software, Resources, Investigation.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used DeepL and Grammarly in order to improve language clarity and grammar. ChatGPT was also used as a language-support and analytical assistance tool. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data availability

Anquela, A. B., Hernandez Zetina, S. L., Verdú Candela, Á., Chiva Gil, J. V., Martínez Montes, C., & Martin, A. (2025). Integrated Datasets for analyses on potentially hazardous locations for women in Valencia, Dublin, San Francisco, and Toluca [Data set]. Zenodo. doi:<https://doi.org/10.5281/zenodo.14726381>

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